Pavan Kolasani

Data Engineer & Data Scientist | SQL, Python, AWS | MSc Computer Science (Data Science), First Class

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Stamp 1G — eligible to work in Ireland immediately

PROFESSIONAL SUMMARY

Data engineer and data scientist working in healthcare data migration at UPMC Ireland, taking data through cleansing, mapping, validation and signoff under strict information governance. AWS Certified Generative AI Developer - Professional and AWS Certified Data Engineer - Associate, with a First Class MSc in Computer Science (Data Science) from TU Dublin. Portfolio work focuses on building models and analyses that hold up under scrutiny: leakage controls, cost-based thresholds, confidence intervals, drift checks and error analysis.

TECHNICAL SKILLS

Languages and querying: SQL (joins, window functions, aggregation, reconciliation queries), Python (pandas, NumPy), DuckDB, R (basic)

Data engineering: ETL/ELT pipelines, source-to-target mapping, data profiling and validation, schema modelling, data migration

Cloud: AWS (Glue, Redshift, S3, Lambda, Athena, Kinesis; generative AI services, Professional certified), Azure (familiar), Databricks (exposure)

ML and evaluation: scikit-learn, XGBoost, TensorFlow, YOLOv8; model evaluation, hypothesis testing, bootstrap confidence intervals, McNemar tests, drift monitoring (PSI)

Analytics and BI: Power BI, Tableau, Excel (Power Query, PivotTables), KPI design, statistical reporting

Engineering practice: Git, GitHub Actions CI, pytest, FastAPI, Docker, technical documentation

PROFESSIONAL EXPERIENCE

EHR Data Migration Specialist, UPMC Ireland, Dublin Aug 2026 - Present

- SQL-based migration of healthcare data between systems, from source cleansing and source-to-target mapping through testing, validation and signoff.
- Data quality and profiling checks: duplicate and NULL handling, referential integrity and source-to-output count reconciliation.
- Work with clinical, IT and vendor teams on mapping decisions, escalating data issues and documenting assumptions under GDPR and information governance.

Junior AI/Data Engineer Intern (Remote), Innovorex Jul 2025 - Sep 2025

- Processed, cleaned and validated 50,000+ structured records for a learning analytics platform.
- Built 3 Python ETL workflows for ingestion, transformation and reporting, cutting manual data preparation by about 40%.
- Defined 5 platform KPIs with product and engineering teams, improving dashboard adoption.

Data and Content Analyst Intern, Crewbytes Jan 2024 - Jun 2024

- Cleaned and standardised datasets of 10,000+ records, reducing inconsistencies by about 35%.
- Documented 20+ recurring anomaly patterns, unblocking the product team's reporting.

Data Analyst Intern, SkillBanc May 2022 - Jul 2022

- SQL-based cleaning and trend analysis for monthly management reviews; 4 Excel dashboards adopted by management.

SELECTED PROJECTS

GenAI Policy Risk Analysis - github.com/kpavan27/genai-policy-risk-analysis

- Mapped the 20 clauses of Google's Generative AI Prohibited Use Policy onto an off-the-shelf moderation signal and measured the gaps on 5,082 real user prompts: 12 clauses had no moderation category; moderation caught 15% of human-labelled toxic prompts and 14% of jailbreaks.

- Excluded moderation-pre-filtered rows to avoid circular evaluation; quantified the review-load cost of lowering the threshold (6x flag volume for 68% recall) and wrote a ranked risk memo.

Email Abuse Detection - github.com/kpavan27/email-abuse-detection

- Spam classifier on 33,716 emails reaching 97.6% recall at a 0.41% false-positive rate, significantly better than a Naive Bayes baseline (McNemar p = 0.004).
- Found and removed label shortcuts (mailbox identity tokens, dates) via feature introspection; de-duplicated and grouped splits by campaign; showed recall falling to 91.7% on newer spam (PSI 0.38). DuckDB SQL analysis, FastAPI service, CI.

Zero-Shot Scene Classification - MSc Dissertation (First Class), TU Dublin

- Interpretable zero-shot classifier (YOLOv8-Large detection, TF-IDF object weighting, L1-normalised scoring): 68.5% Top-1 accuracy on 4,400 Places365 images across 8 unseen categories.

EDUCATION

MSc Computer Science (Data Science) - Technological University Dublin, Ireland 2024–2025
First Class Honours

B.Tech Data Science and Artificial Intelligence - ICFAI Tech University, India 2020–2024
CGPA 8.3/10

CERTIFICATIONS

- AWS Certified Generative AI Developer - Professional - Amazon Web Services, Oct 2026 (validation 1b9fac287b294aee85d34a35ecc5591c)
- AWS Certified Data Engineer - Associate (DEA-C01) - Amazon Web Services, 2026
- Generative AI & LLM Development - DeepLearning.AI, 2024
- AWS Cloud Practitioner Essentials (course) - Amazon Web Services, 2025